

Sleep Scores: Your Smartwatch Is Not a Sleep Laboratory (And That's Fine)

The sleep score should answer “how restorative was last night’s sleep?”, not “am I ready to train?”. In practice this means sleep duration, continuity, regularity, and a small bit of physiology. The important point from the literature is that sleep trackers are usually more reliable for broad sleep-wake behaviour and trends than for exact metrics.

However, people do not naturally read a familiar 0-100 headline as an absence-of-warning scale. They read 100 as exceptional or nearly perfect sleep, 90 as very good sleep, and the distance between 90 and 100 as a meaningful difference in quality. The number therefore has to represent degree of restorativeness, not merely whether the system found a large enough reason to deduct points. A normal, uninterrupted or unremarkable measurement can support a strong result, but it cannot by itself be treated as evidence of exceptional sleep.

Wrist actigraphy (method estimating sleep/wake using movement data from a wrist device) is strong for detecting sleep overall, but weaker for detecting wake, and consumer devices infer stages from movement and cardiorespiratory signals rather than measuring brain activity directly (obviously). [1]

We’ll start by understanding what the mainstream providers currently do to derive their sleep scores. They all follow a common pattern: mixing sleep amount, sleep quality/continuity, and some form of physiological recovery signal, but they differ in how much they expose the logic.

Fitbit’s sleep score has typically been described through dimensions such as duration, restoration, and restlessness, and Google’s more recent redesign appears to be moving toward a more transparent version that shows total sleep, time to reach “sound sleep,” interruptions, and full awakenings, in some cases with comparisons by age and gender.

This is a sensible evolution because users can understand it (one of the bits of the health app I don’t hate). [2]

Apple entered this space later with a Sleep Score in watchOS 26. Public reports say the score uses signals such as heart rate, wrist temperature, blood oxygen, and respiratory data, while other reporting indicates Apple also weights more actionable behavioural features like bedtime consistency and time in bed.

The general direction is relatively clear: the company is treating sleep as both a physiological and a behavioural system. [3]

Garmin’s sleep ecosystem is less centered on one single number and more on the surrounding tools. Public guides emphasise overnight stress, Body Battery, respiration rate, and their Sleep Coach, which adjusts sleep goals based on activity, HRV, and naps.

This serves as a useful reminder that a good sleep product is not just a number in the morning; it is also guidance on how much sleep the person likely needs, we want this number to help the user improve their readiness. [4]

WHOOP leans even harder into the “sleep as recovery fuel” framing. WHOOP directly links

higher strain to higher Sleep Need, and the company's Healthspan work treats sleep consistency and sleep duration as core long-term health metrics. [5]

Moving to the scientific literature, we see 3 main conclusions.

Firstly, sleep quantity and regularity are more robust than fine-grained sleep staging in wearables. Wrist actigraphy is good at identifying sleep versus wake overall, but it has difficulty with wake detection, and stage-level inference remains much less secure.

More advanced wearable staging models using PPG and contextual windows can achieve good performance, but are still not the same as lab PSG (polysomnography - proper lab sleep studies). That doesn't make them useless but it means we should weight their outputs conservatively. [6]

The second is that sleep regularity matters far more than many apps used to admit. Large observational work summarised in recent reports found that irregular sleep schedules were strongly associated with worse cardiovascular outcomes, and other large-scale work linked higher sleep regularity to a lower mortality risk.

From a product-design point of view, this means regularity should not be a tiny bonus metric but rather a major component of the score. [7]

The third is that wearables are most useful when they show trends and behaviour links, not when they pretend to be medical sleep labs. Sleep specialists quoted in recent reports make this point repeatedly: these devices can help people identify patterns around alcohol, late eating, bedtime drift, illness, and disturbance, but exact stage numbers should not be treated as truth. [8]

The more important question for us is not just "are wearables imperfect?", because they obviously are. The more useful question is which parts of sleep are reliable enough and important enough to drive a 'sleep score'. This is where I think the feature order becomes much clearer.

For newer products such as the Fitbit Air, we should be more optimistic than a generic "wrist trackers are bad" argument. Scientific evaluation videos compare the device against stronger reference measurements and suggest that these devices perform well enough to be useful, even if not perfect.

The important distinction is that this should increase our confidence in wearables as a practical data source, but it should not change the basic scientific hierarchy. Even a good wearable is still more trustworthy for sleep duration, timing, continuity and broad sleep behaviour than it is for exact REM/deep/light stage breakdowns. [9]

The strongest order of importance looks like this:

- Sleep duration: This should likely be the largest part of the score. If the user does not get enough total sleep, the rest of the score should not be able to fully rescue it. Sleep stages can look "good" within a short sleep, but 4 hours of sleep is still, at the end of the day, 4 hours of sleep.
- Sleep regularity: This is the most underweighted feature in many current apps. Large cohort work has found sleep regularity to be strongly associated with mortality and cardiovascular outcomes, in some cases more strongly than just duration alone. From a scoring perspective, this means consistency should not be a tiny bonus. It should be a major block of the score. [10] [11]
- Continuity and awakenings: Sleep that is constantly interrupted is not the same as one continuous block of sleep. This is where wake after sleep onset, number of awakenings, restlessness, and sleep efficiency matter. Wearables are not perfect at wake detection, but this is still more reliable and actionable than obsessing over the exact stage percentages.

- Timing and sleep onset: Bedtime drift, very late sleep, and taking a long time to fall asleep should also matter, but slightly less than duration, regularity and continuity. Timing is important because it reflects circadian stability, not just “when did I happen to go to bed”.
- Overnight physiology: HRV, RHR, respiratory rate, SpO₂ and temperature can help explain why sleep may have been poor or why the body looked stressed overnight. However, these should be smaller components of the sleep score than they are in a readiness score. They are more useful as context than as the definition of the sleep’s quality.
- Sleep stages: REM and deep sleep should be included, but carefully. There is evidence that longer deep sleep, higher sleep efficiency and earlier REM latency are associated with more favourable wearable-derived health markers, but stage detection is still less secure than duration and timing. The key point is: sleep stages are useful supporting evidence, not the backbone of our score. [12] [13]

Given this, the behavioural core should weight duration at 45%, regularity at 20%, continuity at 25%, and timing and sleep onset at 10%. Valid sleep stages should then account for 5% of the combined sleep score. Overnight physiology should act as a small downward modifier after that calculation rather than supply a permanent block of positive points.

These weights should be fixed for the initial algorithm. Future validation can test whether the stage weight should vary by provider, but measurement unreliability is the reason stages receive only 5%; it does not make REM and N3 biologically unimportant. Physiology should always have less influence than duration, regularity and continuity.

The sleep score should be partly personalised, although not in the same way as readiness or strain. Duration should use the personalised core sleep need defined in Sleep Need, Debt, Recovery and Stages. Regularity should be strongly personalised because the important question is whether the user is consistent relative to their own life. Continuity should also be compared with the person’s baseline because some people naturally wake more often than others.

The score should also have a confidence layer. If duration and timing are available but stages are missing, the software should still calculate a sleep score as the strongest features are still present. If sleep stages look strange but duration, regularity and continuity are good, the score shouldn’t collapse.

If duration is short, bedtime is late, and awakenings are high, the score should drop even if the device claims a decent amount of REM. This is the core principle: the sleep score should reward the most reliable and most important sleep features first.

This gives us a sleep score that answers the right question: “how restorative was last night’s sleep?”, rather than pretending the wearable is a medical sleep lab or accidentally turning the sleep score into a readiness score.

These principles describe what the sleep score should value. To turn them into an implementable algorithm, each component also needs a defined 0-100 scale, a calculation method and clear rules for missing or unreliable data.

Across the core components, 100 should be reserved for an exceptionally favourable observed result: sleep need was fully met, continuity was nearly complete, and schedule behaviour was very closely aligned. A value around 80 represents a good but clearly improvable result, around 60 indicates a meaningful problem, and values below 40 should be reserved for a large shortfall or severe disruption. This matters because “nothing triggered a warning” is not the same claim as “this was exceptional sleep”. Straight-line interpolation should be used between the anchor values described below.

Calculating sleep duration

Sleep duration accounts for 45% of the score. Sleep achieved comes from the valid 24-hour sleep day defined in the accompanying sleep paper, while the comparison target is core sleep need rather than the larger recovery opportunity. Recovery opportunity is advice about how much time to allow for sleep; failing to complete all of it should not manufacture a new debt on top of the existing one.

The duration calculation should use the complete shortfall against core sleep need. A small miss is handled by the gentle upper part of the curve rather than being erased before scoring:

$$\text{duration shortfall} = \max(0, \text{core sleep need} - \text{valid sleep achieved})$$

The duration component should then use these anchor points, with straight-line interpolation between them:

- 0 minutes of shortfall: 100.
- 15 minutes: 97.
- 30 minutes: 94.
- 60 minutes: 84.
- 90 minutes: 70.
- 120 minutes: 55.
- 180 minutes: 30.
- 240 minutes: 10.
- 300 minutes or more: 0.

This keeps a 15 or 30-minute miss in the very-good range while ensuring that an hour or 2 of missing sleep is visible. A night that is 3 or 4 hours short cannot hide behind good-looking continuity or stage data. Sleep above core need should not lose points in this component.

A night more than 2 hours above need can be marked as unusually long and explained alongside illness or recent restriction, but long sleep is not automatically bad sleep. The companion paper already explains why extra sleep can't create a positive debt balance for later.

Calculating sleep regularity

Sleep regularity should be understood as a short rolling behaviour rather than a component contained entirely inside one night. The Personal Metric Baselines paper explains why bedtime and wake time need circular calculations, separate workday and free-day references, and a stable device source.

Where a full sleep-wake timeline is available, the Sleep Regularity Index (SRI) is useful for the longer view because it asks how often the person is in the same sleep or wake state at points 24 hours apart. It is sensitive to naps, awakenings and all-nighters, while a simple standard deviation of bedtime is not.

Composite Phase Deviation is useful when the focus is how much the latest sleep midpoint differs from both the previous day and the person's usual timing. These measures answer slightly different questions, so we should not casually average every regularity metric we can calculate. [15]

These measures are useful for longer-term reporting, although the nightly component needs a simpler and more interpretable calculation. It should use the actual bedtime and wake-time drift from the correct day-type reference. Calculate their circular differences, then take the mean:

$$\text{schedule drift} = \frac{|\text{bedtime drift}| + |\text{wake-time drift}|}{2}$$

The regularity score should use these anchors:

- 0 to 15 minutes of schedule drift: 100.
- 30 minutes: 95.
- 60 minutes: 85.
- 90 minutes: 70.
- 120 minutes: 50.
- 180 minutes: 20.
- 240 minutes or more: 0.

Straight-line interpolation fills the gaps. At least 7 valid comparable nights are needed before this personal component can be treated as established, with 28 preferred. Before then, the latest bedtime and wake time can still be shown, but the regularity component is provisional.

The 7-day SRI should be reported beside the component as context instead of being used as a second penalty for the same behaviour. An SRI of 100 means perfectly repeated sleep-wake timing, although cohort cutoffs should not be treated as universal nightly score boundaries. [21]

Calculating continuity

Continuity should focus on what happened after the sleep session actually began. This keeps it separate from sleep onset latency. The main measurement is wake after sleep onset (WASO), supported by the number of awakenings lasting more than 5 minutes.

Sleep efficiency should be shown, but it should not be averaged as another independent input because efficiency, WASO and awakenings are largely different descriptions of the same lost sleep.

Expert consensus judged WASO of 20 minutes or less as a sign of good sleep quality across the lifespan. For most adult groups, 41 to 50 minutes was already considered poor, with 51 minutes or more clearly outside good sleep quality.

Sleep efficiency of at least 85% was considered good, while 74% or lower was poor for most age groups. The same review found 0 or 1 awakening lasting more than 5 minutes was good, while 4 or more was poor; 2 awakenings remained acceptable for older adults. [14]

WASO can therefore drive the continuity score using these anchors:

- 0 to 10 minutes: 100.
- 20 minutes: 95.
- 30 minutes: 85.
- 40 minutes: 70.
- 50 minutes: 50.
- 60 minutes: 30.
- 90 minutes or more: 0.

The score is then capped when the supporting measurements show a clearly fragmented night. Maintenance efficiency below 85% caps continuity at 65, below 75% caps it at 30, and below 65% caps it at 15. Four or more awakenings lasting over 5 minutes cap it at 45.

3 such awakenings cap it at 70 and 2 cap it at 85, except that 2 awakenings produce no cap for somebody aged 65+. These are caps rather than separately weighted penalties, which stops the same wake time being counted 3 times.

Wearables are often weakest at detecting quiet wake, so provider-specific personal comparison still matters. A night that is more than 2 spreads worse than the person's 28-night WASO or maintenance-efficiency reference should be flagged even if it narrowly remains inside a population anchor.

The population thresholds give the curve a sensible shape; the personal reference tells us when the night was unusual for this user. Normal polysomnography also changes with age: across adult decades, sleep efficiency falls and WASO rises, so older sleep should not be judged as though it came from a healthy 20-year-old. [16]

Calculating timing and sleep onset

Timing and sleep onset account for 10% of the score. This component has 2 possible inputs. Sleep latency measures the time from trying to sleep to the first sustained sleep.

Circadian alignment measures how far the sleep midpoint sits from the person's chosen or learned target window. Regularity asks whether the schedule repeats; alignment asks whether the schedule happened near the window that normally fits the person's life and daytime function.

The latency subscore should use the consensus ranges from the sleep-quality review:

- 0 to 20 minutes: 100.
- 30 minutes: 95.
- 45 minutes: 75.
- 60 minutes: 40.
- 90 minutes or more: 0.

A very short latency should not be penalised here. It may reflect high sleep pressure, but duration and recent deficit already describe that situation more directly.

The second part of this component is circadian alignment. This should compare actual midsleep with the midpoint of the target sleep window using circular time:

- Within 15 minutes: 100.
- 30 minutes: 95.
- 60 minutes: 85.
- 90 minutes: 70.
- 120 minutes: 50.
- 180 minutes or more: 0.

When both values are available they can be averaged equally. If the device only supplies detected sleep start, latency remains missing and alignment carries the component. When mid-sleep alignment is derived from the same learned bedtime and wake-time reference as regularity and produces the same underlying drift, it should not be scored as a second independent input. In that case timing uses latency alone; if latency is also missing, the timing weight is rescaled across the other core components. There should be no universal penalty simply for sleeping between 2am and 5am.

That may be badly aligned for one person and completely normal for a night worker. Travel and timezone shifts should reduce confidence until a stable local pattern returns.

Calculating overnight physiology

Overnight physiology should behave as supporting evidence. HRV, RHR, respiratory rate, oxygen saturation and temperature are valuable for understanding the night, but they do not directly measure whether the person was asleep or whether that sleep was continuous.

Reviews also find that objective sleep measurements explain only part of how people rate their own sleep, with results varying substantially between studies. [18]

The component should begin at 100 and only move down when comparable overnight signals leave the personal ranges defined in the accompanying Personal Metric Baselines paper. The signals can be grouped so correlated measurements do not each pretend to be independent:

- Autonomic group: HRV and resting heart rate.
- Respiratory group: respiratory rate and oxygen saturation.
- Temperature group: overnight temperature variation.

One abnormal group gives a physiology score of 80, 2 groups give 60, and all 3 give 40. A group becomes abnormal when one signal crosses its metric-specific alert boundary on 2 consecutive nights, or when 2 related signals cross together on the current night.

The physiology score becomes a modifier: 100 subtracts 0 points, 80 subtracts 3, 60 subtracts 6, and 40 subtracts 9. A severe absolute oxygen-saturation reading or a provider quality warning can create an immediate respiratory flag, although this remains a wellness warning rather than a diagnosis. Missing physiology stays missing and creates no point deduction. It cannot add points or rescue poor sleep.

Calculating sleep stages

Sleep stages account for 5% of the combined score when valid. Their small weight reflects measurement uncertainty in consumer wearables, not a claim that sleep architecture is unimportant. The companion paper defines the validity checks: at least 90% timeline coverage, agreement with total sleep, valid non-overlapping intervals, a stable provider and algorithm source, and enough nights for a personal reference. These checks should be applied before the stage component is calculated. Once the data passes them, the stage score can be derived from personal deviation and broad age-based plausibility limits.

Population stage values are best used as plausibility rails. They change with age and they come from polysomnography rather than a wrist classifier. Across adulthood, N1 (light sleep) and N2 (stable light sleep) generally rise while slow-wave and REM sleep fall, and healthy adult sleep efficiency also declines with age. [16] [19]

The sleep-quality consensus found REM around 21% to 30% and N3 (deep sleep) around 16% to 20% compatible with good sleep in adults aged 26 to 64. It found N3 around 20% to 25% compatible with good sleep in teenagers.

It also found that REM at 41% or more was not a good-quality marker in young adults, adults or older adults, and N3 of 5% or less was not a good marker from the teenage years through adulthood. The panel did not agree on many positive stage ranges, particularly in older adults, which is a useful warning against building a detailed age quota table from weak evidence. [14]

The stage component should therefore compare REM and N3 percentages with the provider-specific 28-night personal reference first. We can use the centre and spread from the baseline paper:

- Both stages within 2 spreads of their references: 100.
- One stage between 2 and 3 spreads away: 80.
- One stage beyond 3 spreads, or both stages beyond 2 spreads: 60.
- The same stage remains beyond 2 spreads for 3 consecutive valid nights: 40.

Age plausibility acts as a cap. For ages 13 to 17, REM at 10% or less or N3 at 5% or less caps the stage component at 60. For ages 18 to 64, REM at 41% or more or N3 at 5% or less also caps it at 60.

For ages 65 and older, REM at 41% or more caps it at 60, while there is no hard low-N3 cap because the consensus evidence was not strong enough to set one. An impossible or internally inconsistent stage result should have failed the earlier quality checks and remain missing.

These are broad guardrails, not targets to chase. A teenager does not fail sleep just because N3 was 18% instead of 20%, and an adult does not receive bonus points for forcing REM into a narrow band.

Valid stages enter as a direct 5% component. A stage score of 100 contributes the full 5 points, 80 contributes 4, 60 contributes 3, and 40 contributes 2. Because the behavioural core supplies the other 95%, even a large stage deviation cannot overwhelm duration, regularity or continuity. Missing stages remain missing and are never represented numerically as excellent stage evidence.

Combining the components

The principles above can be written as the following calculation:

$$\text{behavioural core} = 0.45D + 0.20R + 0.25C + 0.10T$$

When stages are valid:

$$\text{stage-adjusted score} = 0.95 \text{ behavioural core} + 0.05 \text{ stage score}$$

When stages are missing, the stage-adjusted score equals the behavioural core. The missing 5% is neutralised across the complete behavioural result rather than assigned to one other component or represented as a stage score of 100.

$$\text{sleep score} = \text{clamp}(\text{stage-adjusted score} - \text{physiology modifier}, 0, 100)$$

Duration is required. At least 2 of regularity, continuity and timing/onset must also be available; otherwise a complete sleep score should be withheld. If one core component is unavailable, the available core weights are rescaled to 100%.

Missing optional evidence never enters the arithmetic as 100. If either physiology or stages is missing, the final score is capped at 97; if both are missing, it is capped at 95. These are evidence-sufficiency caps rather than deductions: a good night can still score highly, while an exceptional claim requires enough measured evidence to support it.

Some nights also need caps. A duration shortfall of 3 hours caps the final sleep score at 45, 4 hours caps it at 30, and 5 hours or more caps it at 20. WASO of 90 minutes or more, maintenance efficiency below 65%, or 4 or more long awakenings caps the final score at 50. These caps preserve the principle established above: an attractive stage estimate cannot rescue a very short or severely fragmented night.

Missing data and confidence

Confidence should sit beside the result and never be multiplied into the score. High confidence requires a valid main sleep, at least 90% sleep-period coverage, all 4 core components, a stable provider source and a 28-night personal reference. Moderate confidence has the essential sleep record and at least 3 core components, but only 7 to 27 comparable nights or missing optional data. The high-end evidence caps above prevent incomplete optional coverage from supporting an exceptional headline while preserving a separate, explicit confidence label.

Low confidence means the score had to use sparse history, a recent device change, incomplete sleep intervals or only the minimum 3 core components. Missing duration, an invalid main-sleep session or fewer than 2 supporting core components means no score.

Interpreting the score

The following bands preserve the ordinary meaning people assign to a 0-100 quality score:

- 95-100: exceptional sleep. The reliable core measurements were favourable across the night, with enough evidence to support an exceptional result.

- 85-94: very good sleep. The night was strongly restorative, with limited room for improvement.
- 75-84: good sleep. The night was broadly restorative, although at least one meaningful dimension was not ideal.
- 60-74: compromised sleep. A clear shortfall or disruption is likely to matter the next day.
- 40-59: poor sleep. One large problem or several agreeing problems materially reduced restorativeness.
- 0-39: very poor sleep. The night contained severe restriction, fragmentation or several major problems.

The component explanation should remain visible. A score of 68 caused by short sleep calls for a different response from 68 caused by severe fragmentation or schedule disruption.

Conclusion

Taken together, the sleep score supported by this research has the following form:

- Duration accounts for 45% and is scored from the full shortfall against core sleep need. A 1-hour shortfall scores 84, 2 hours scores 55, 3 hours scores 30 and 5 hours scores 0.
- Regularity accounts for 20% and uses circular bedtime and wake-time drift from the correct personal day-type reference. Drift up to 15 minutes scores 100, 90 minutes scores 70 and 3 hours scores 20.
- Continuity accounts for 25% and is driven by WASO. Up to 10 minutes scores 100, 40 minutes scores 70 and 90 minutes scores 0, with maintenance efficiency and long awakenings used as caps.
- Timing and sleep onset account for 10%. Latency up to 20 minutes scores 100, 60 minutes scores 40 and 90 minutes scores 0. Midsleep alignment uses a separate personal target window, carries no universal late-night penalty and is omitted when it duplicates the regularity input.
- Overnight physiology is a 0, 3, 6 or 9-point grouped personal-baseline modifier. It cannot improve the score and missing physiology creates no deduction.
- Valid sleep stages account for 5% of the combined score after the quality rules in the companion paper pass. They use a provider-specific 28-night reference with broad age plausibility caps, never REM or deep-sleep debt. Missing stages remain missing and the behavioural core carries the score subject to the evidence cap.
- Duration is mandatory. Missing optional inputs remain unscored and cap exceptional claims at 97 or 95, severe duration and fragmentation apply explicit caps, and confidence is reported beside the score.

This produces a sleep score in which 100 requires exceptional measured sleep rather than the mere absence of an alert. Every component has a defined role and no less reliable measurement can overwhelm the core sleep features. The exact interpolation curves and caps

are design values built around the strongest thresholds available, so they should be validated against morning sleep quality and daytime function.

The remaining uncertainty is explicit. Some values come directly from consensus research, others depend on personal baselines, and the interpolation curves and caps are initial implementation choices.

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